

The diagram illustrates the Routing Decision Space, which is divided into three distinct regions based on feature matching quality. Each region contains two pairs of feature tracks, each pair represented by a colored box with its SIFT score, Distance (D), and classification.

- LoFTR Region (Pink background):**
 - Pair 5:** $S=0.71, D=0.38$, Cross-track
 - Pair 6:** $S=0.45, D=0.62$, Cross-track
- LightGlue Region (Orange background):**
 - Pair 3:** $S=0.48, D=0.32$, Cross-track
 - Pair 4:** $S=0.52, D=0.41$, Cross-track
- SIFT Region (Blue background):**
 - Pair 1:** $S=0.28, D=0.12$, Same-orbit
 - Pair 2:** $S=0.32, D=0.18$, Same-orbit

At the top, the overall routing decision is summarized: $S \geq 0.65 \rightarrow R = 0.55$ (Sparse) and $0.25 \leq S < 0.65 \rightarrow R = 0.55$ (Gap).

$S \geq 0.65$ or $D \geq 0.55$

Sparse Pair 5 ion gap

Pair 6
(S=0.45, D=0.62)
Cross-track

Pair 3

Pair 4
(S=0.52, D=0.41)
Cross-track

Pair 1

Pair 2
($S=0.32$, $D=0.18$)
Same-orbit